### 1.4.10 Short Circuit

In a circuit, if the two terminals of a supply are connected directly without load, then the circuit is said to be short circuit. Figure 1.15 shows a circuit in a short circuit condition.

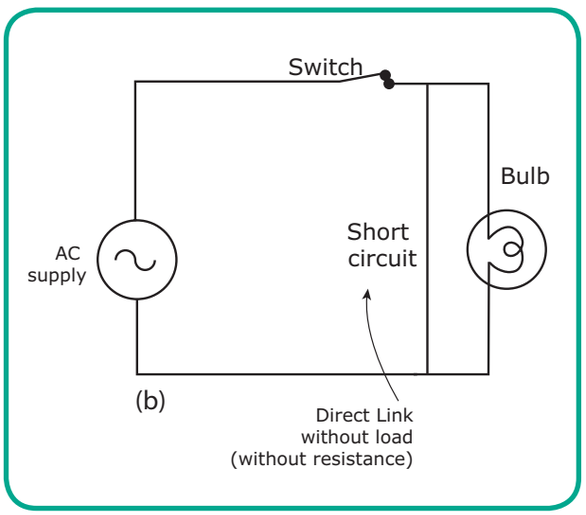



Figure 1.15 Short circuit

In this circuit, the flow of current is infinite $I = \frac{V}{0} = \infty A$. When short-circuit occurs, enormous amount of heat will be produced and heavy spark will occur. This will cause electrical fire accidents. Hence, it is very dangerous. To avoid this, fuse wire, main circuit breakers (MCB) and Earth leakage circuit breakers (ELCB) are used in electrical circuits.

### 1.4.11 Fuse Wire

To avoid fire accidents due to short circuit, thin tin coated copper wires with low melting point is used in series with the circuit. If short-circuit or overload occurs, heat increases in the wires and fuse wire is blown within a second. So, the short-circuit becomes open-circuit, no further current flow and thus avoids fire accident. Figure 1.16 shows the fuse wire connected in an electrical circuit.

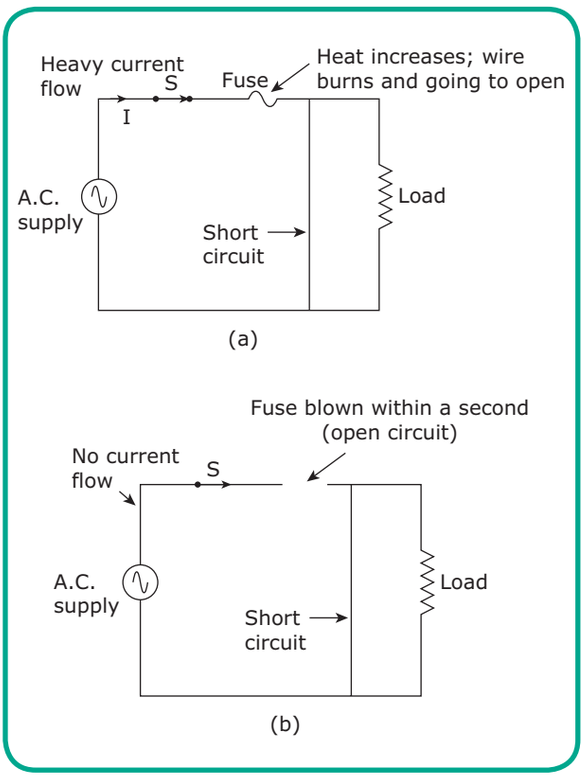



Figure 1.16 Circuit using fuse wire

Fuse wires are available in amperes such as 1A, 2A, 5A and 10A in the market. Presently, fuse wires are substituted with mains circuit breaker and Earth leakage circuit breaker (ELCB).

**Thiruvalluvar referred regarding over load in Thirukkural well before 2000 years.**

**பீலிபெய் சாகாடும் அச்சிறும் அப்பண்டஞ்**
**சால மிகுத்துப் பெயின் #குறள்:475**

**பொழிப்புரை : மயிலிறகு ஏற்றிய வண்டியே ஆனாலும், அந்தப்பண்டமும்அளவோடு ஏற்றாமல் அளவுகடந்து மிகுதியாக ஏற்றினால்அச்சு முறியும்.**

**With peacock feathers light, you load the wain;**

**Yet, heaped too high, the axle snaps in twain.**

**The axle tree of a bandy, loaded only with peacocks' feathers will break, if it be greatly overloaded.**

### 1.4.12 Partial Short Circuit

If short-circuit occurs in few windings of a coil or a transformer, it is called partial short-circuit. Due to this phenomenon,



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